SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE CODE: CSA14 **COURSE NAME: COMPILER DESIGN**

Questions

Annexure A

SHORT ANSWER TEST

Q1. A lexical analyzer produces the token stream:
id + id * id
Explain how the parser validates the syntactic structure of the expression.

Q2. Given the grammar:
S → aSb | ε
Determine whether the string aaabbb belongs to the language using derivation.

Q3. Remove left recursion from the grammar:
E → E + T | T

Q4. Find FIRST(A) for the grammar:
A → aB | ε
B → b

Q5. Determine whether the grammar is suitable for top-down parsing:
A → Aa | b

Q6. Apply left factoring to the grammar:
S → while E do S | while E do S else S

Q7. Construct a parse tree for the expression:
id * id + id
using the grammar:
E → E + E | E * E | id

Q8. Write the recursive descent procedures for the grammar:
S → aA
A → bA | c

Q9. Given the grammar:
S → AB
A → a | ε
B → b
Construct the predictive parsing table.

Q10. Demonstrate the first three actions of a shift-reduce parser for:
id + id

Q11. Identify the handle in the expression:
id * id + id
for the grammar:
E → E + E | E * E | i

Q12. Construct the operator precedence relations for the operators:
+, *
where * has higher precedence than +.

Q13. Explain how operator precedence parsing handles the expression:
id + id * id

Q14. What are LR parsers? Illustrate with a simple parsing scenario.

Q15. Given the grammar:
S → AA
A → aA | b
Write the steps involved in constructing an SLR parsing table.

Q16. Compute FOLLOW(A) for the grammar:
S → AB
A → a | ε
B → b

Q17. Construct the augmented grammar for:
S → aA
A → b

Q18. Find the closure of the LR(0) item:
S' → •S
for the grammar:
S → aA
A → b

Q19. Check whether the grammar is ambiguous:
E → E + E | id
using the string:
id + id + id

Q20. Construct a leftmost derivation for:
id + id * id
using:
E → E + E | E * E | id

Q21. Apply left factoring to:
S → aBc | aBd | b

Q22. Find FIRST(F) for:
F → (E) | id

Q23. Determine whether the grammar is suitable for predictive parsing:
S → aA | aB
A → b
B → c
Justify your answer.

Q24. Construct recursive descent parser procedures for:
S → abS | ε

Q25. Build the predictive parsing table for:

$S \rightarrow aS \mid b$
and show the table entries for terminals a, b, and $.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand